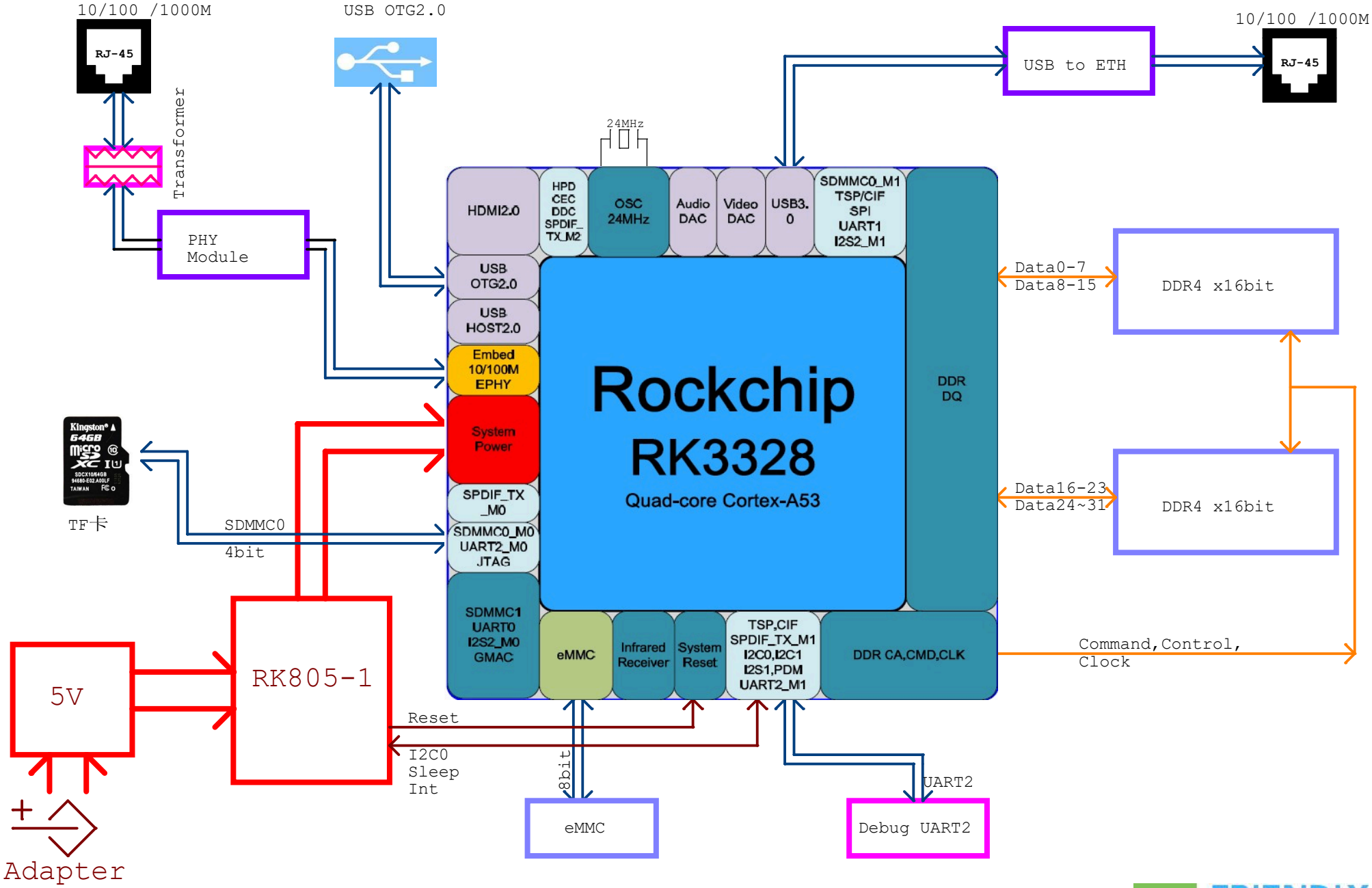
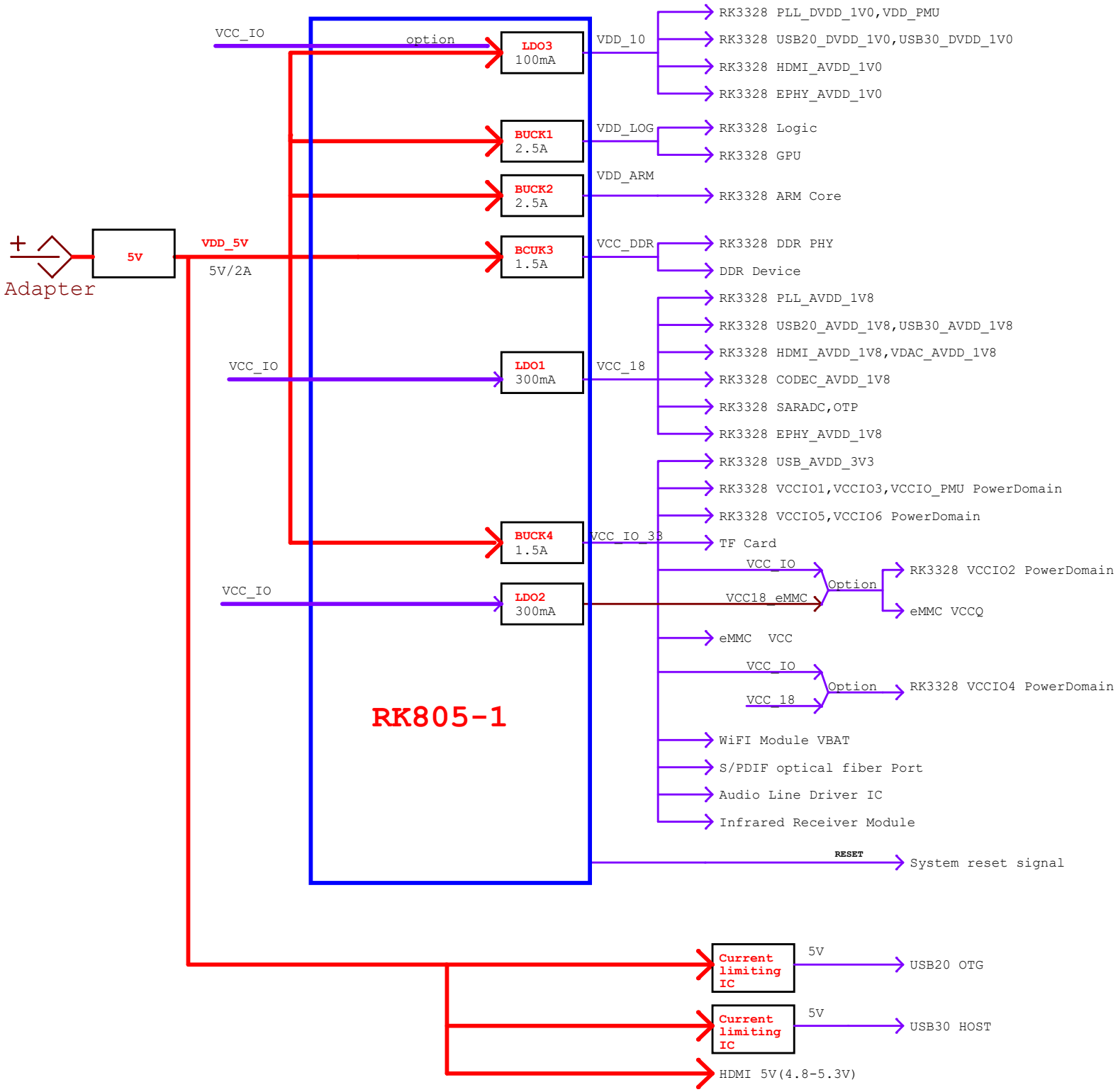


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NanoPi R2S		
Size A3	Document Number Title	Rev 1912
Date: Tuesday, December 17, 2019	Sheet 1 of 14	



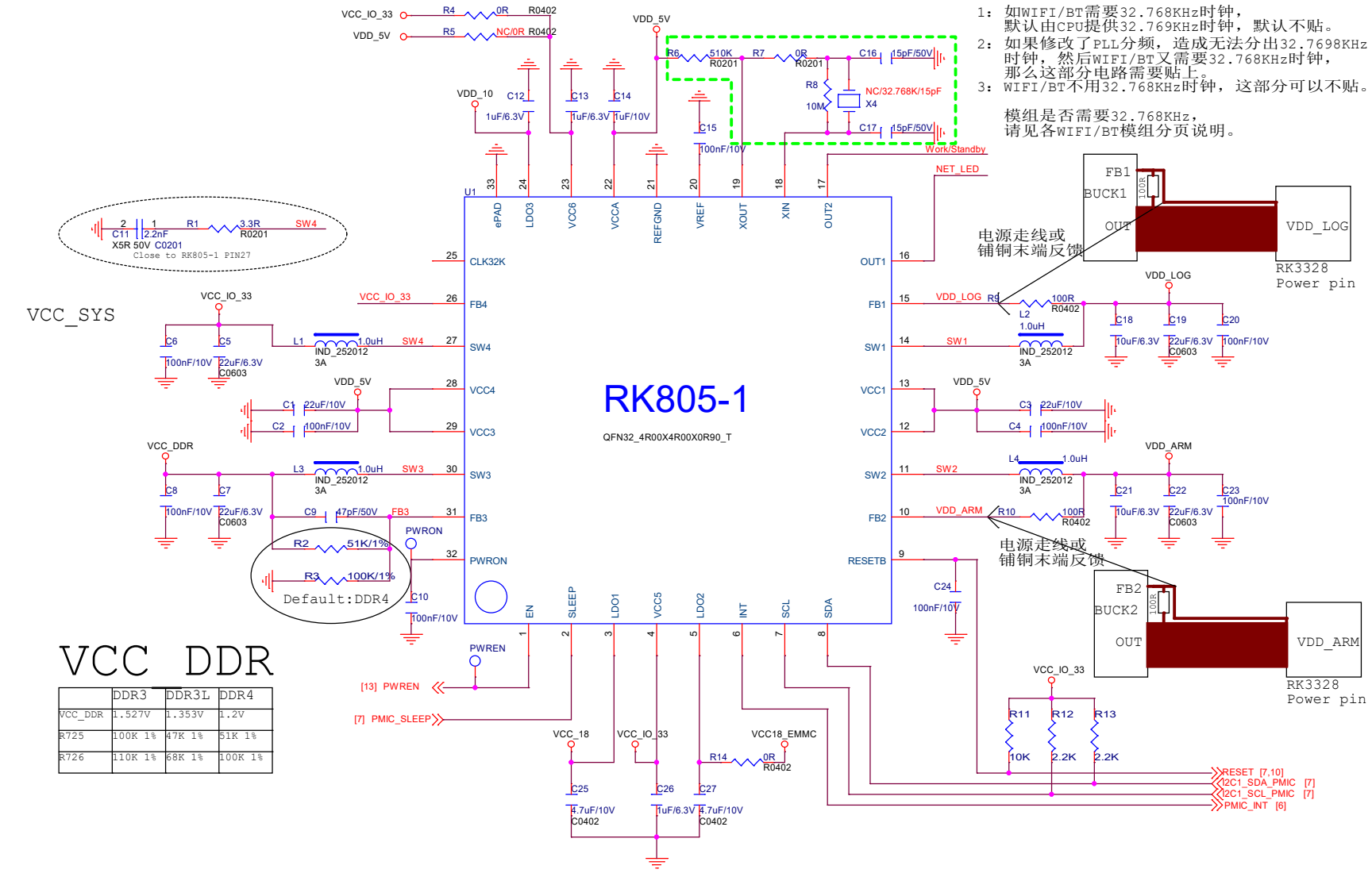
Power Timing

TYPE	VOLT	STEP	PowerName
BUCK1	1.1V		VDD_LOG
BUCK2	1.1V		VDD_ARM
BUCK3	FB=0.8V		VCC_DDR
BUCK4	3.3V		VCC_IO
LDO1	1.8V		VCC_18
LDO2	1.8V		VCC18_eMMC
LDO3	1.0V		VDD_10

IO电源域电压	IO电压＝3.3V	IO电压＝1.8V
VCCIO_PMU	3.3V	不支持
VCCIO1	3.3V	不支持
VCCIO2		1.8V（默认） eMMC
VCCIO3	3.3V（默认）	
VCCIO4	3.3V（默认） WIFI:RTL8723BS	
VCCIO5	3.3V（默认）	
VCCIO6	3.3V（默认）	

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NanoPi R2S		
Size A2	Document Number 02.Power tree-RK805-1	Rev 1912
Date: Tuesday, December 17, 2019	Sheet 2 of 14	

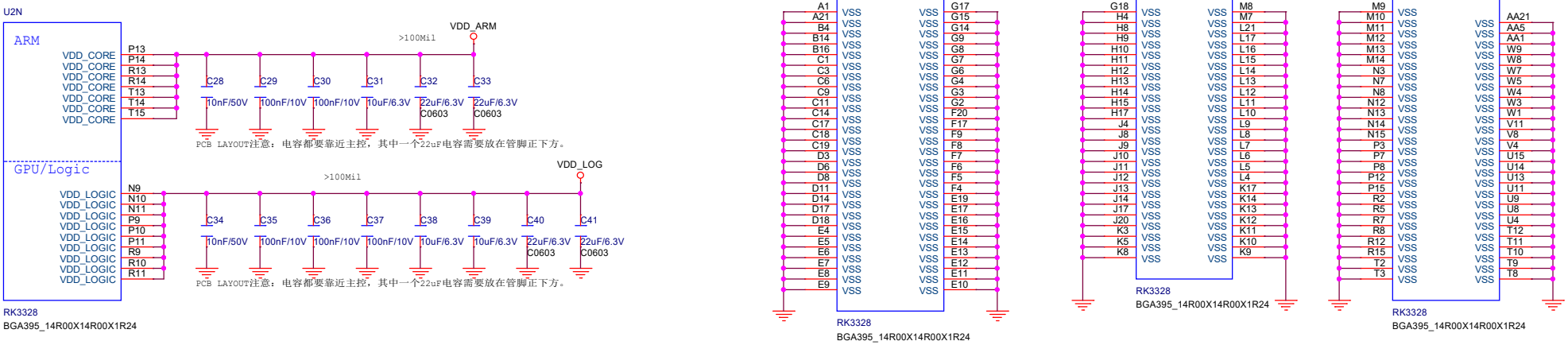


	DDR3	DDR3L	DDR4
VCC_DDR	1.527V	1.353V	1.2V
R725	100K 1%	47K 1%	51K 1%
R726	110K 1%	68K 1%	100K 1%

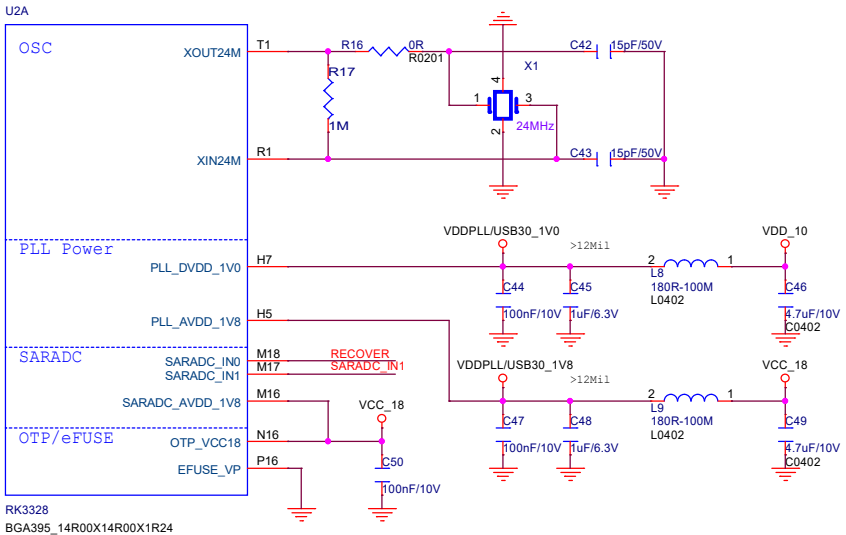
GND GND

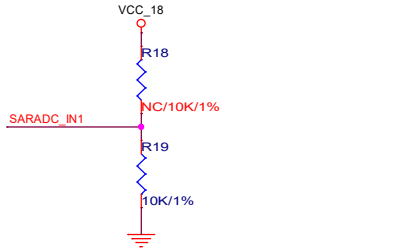
FRIENDLY ELEC

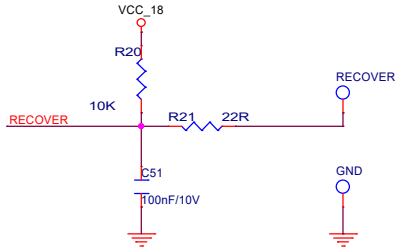
NanoPi R2S		
Size A3	Document Number 03.PMIC RK805-1	Rev 1912
Date: Tuesday, December 17, 2019	Sheet 3 of 14	



RK3328 OSC/PLL/OTP/SARADC

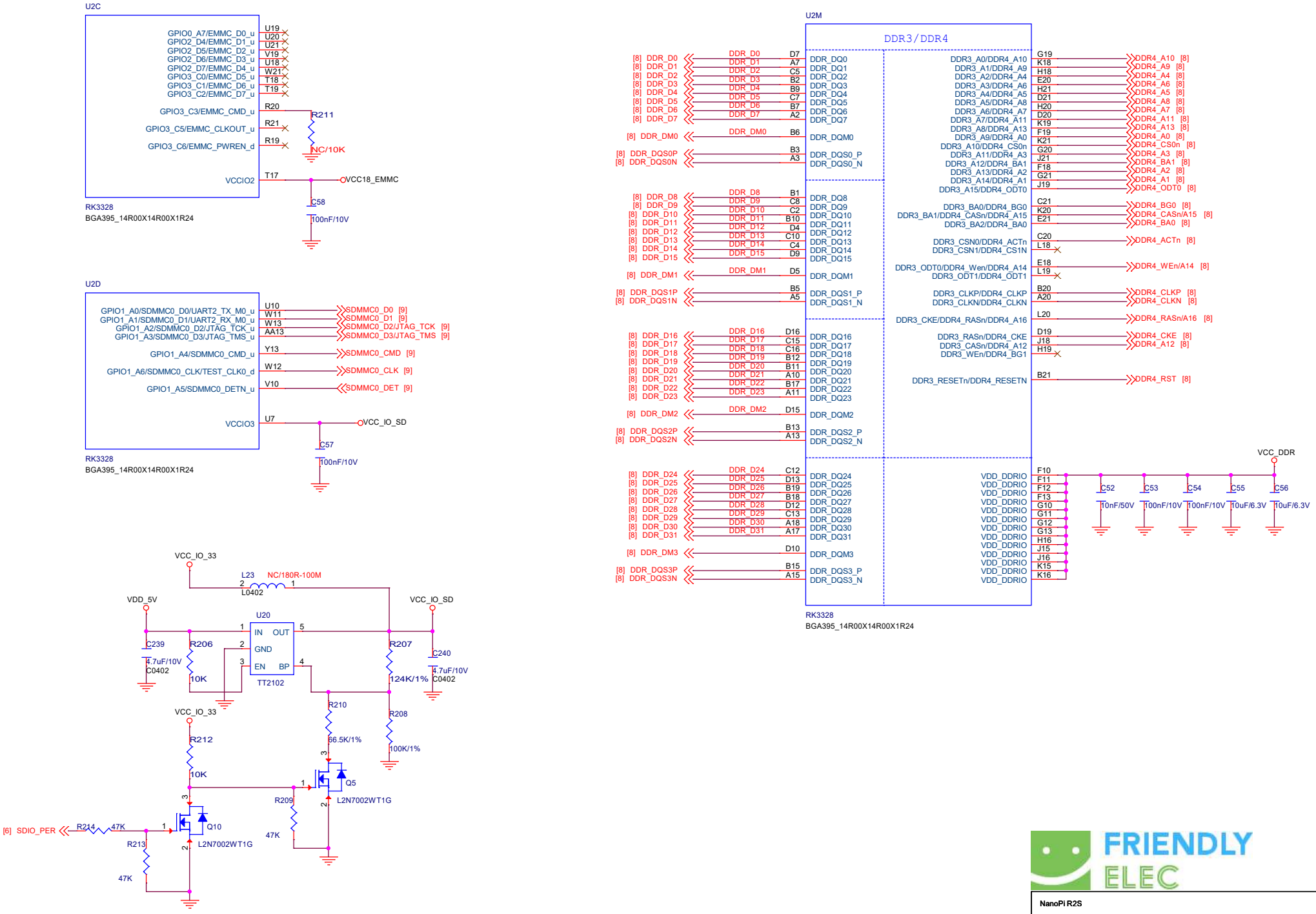


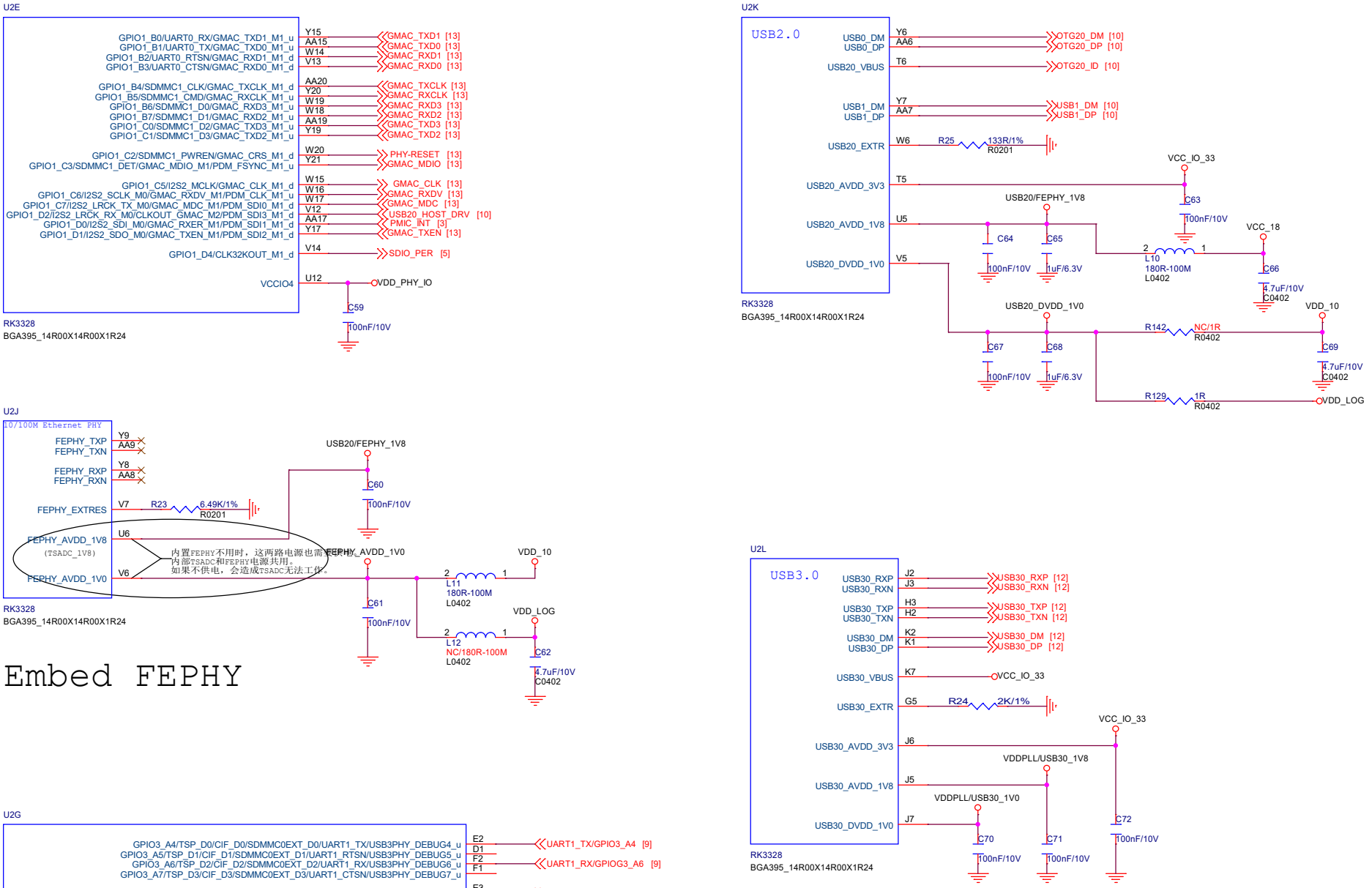


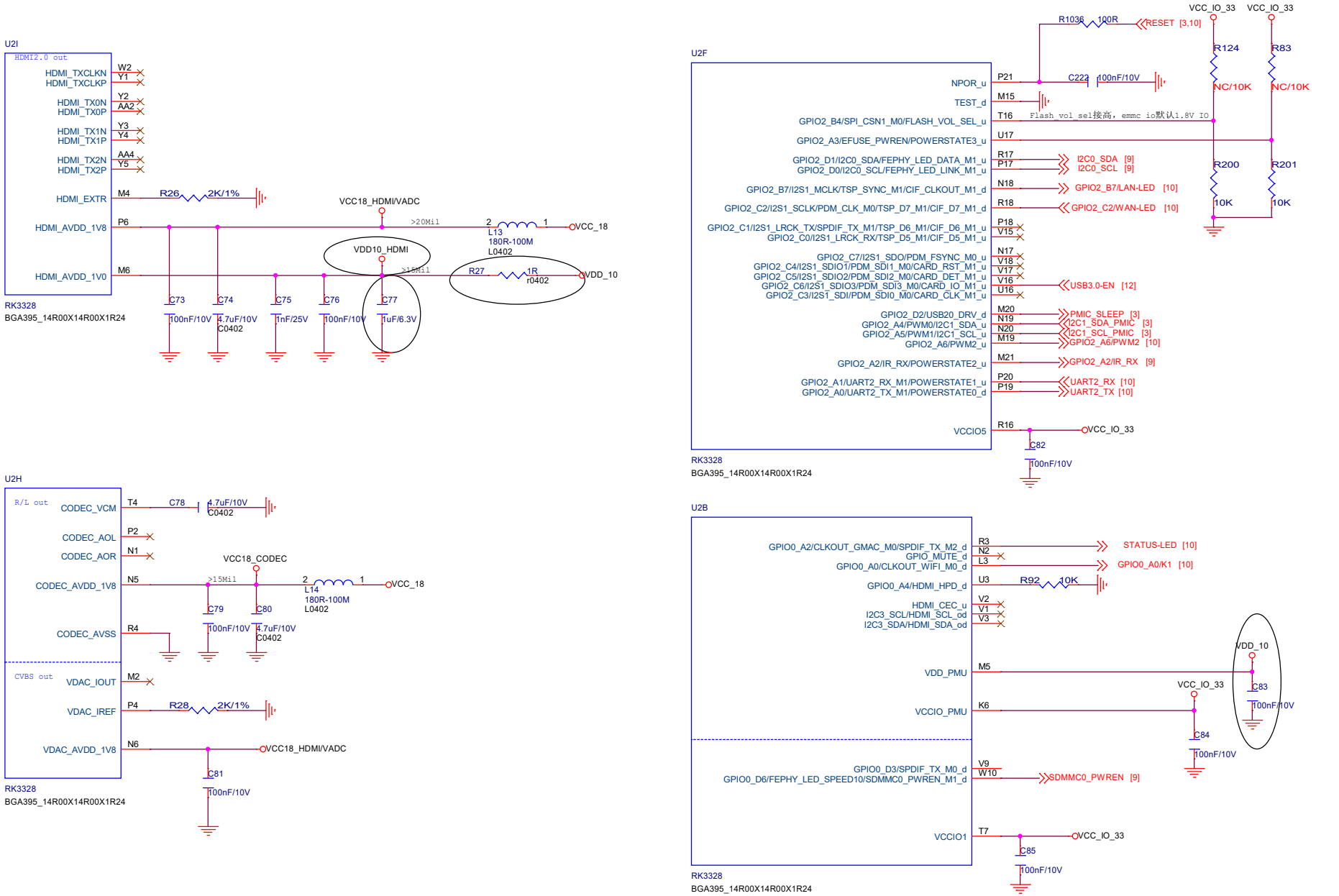


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NanoPi R2S		
Size A3	Document Number 04.RK3328 Power	Rev 1912
Date: Tuesday, December 17, 2019	Sheet 4 of 14	

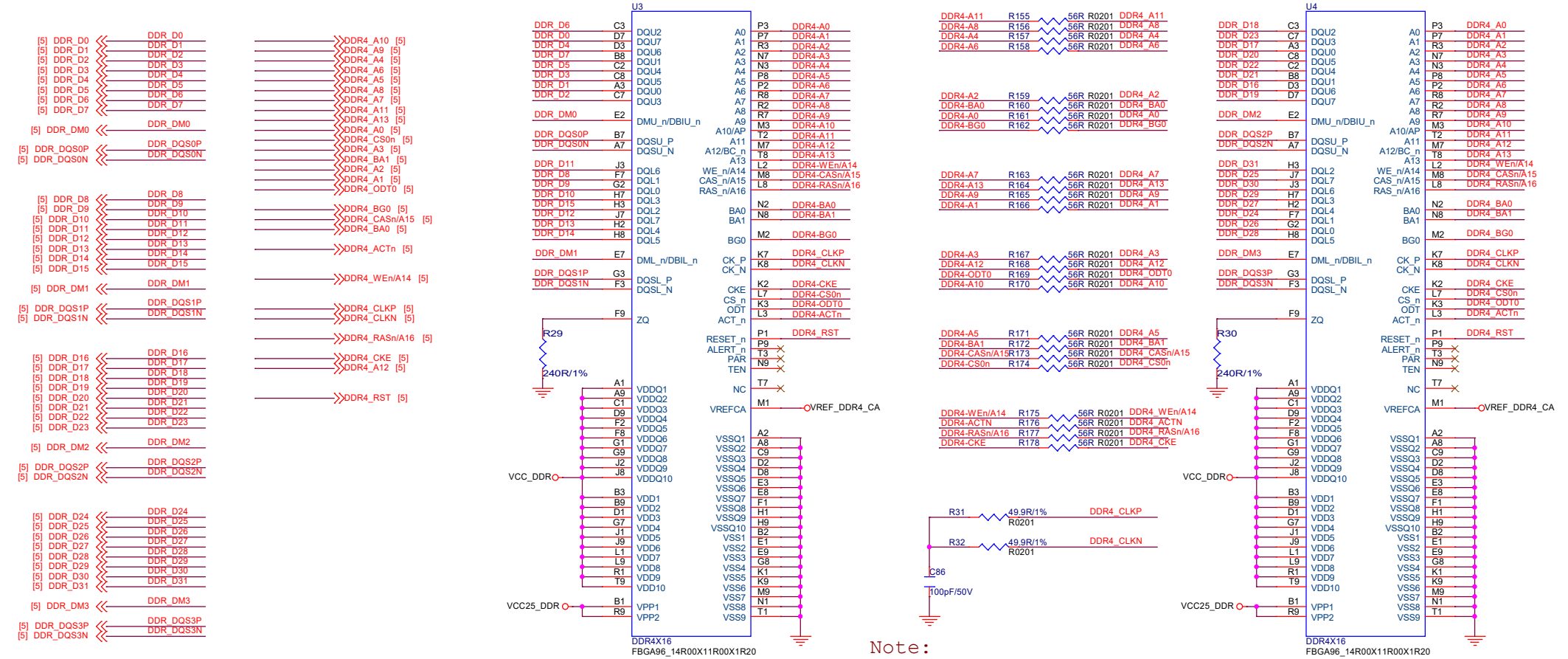

U2C
GPIO0_A7/EMMC_D0_u
GPIO2_D4/EMMC_D1_u
GPIO2_D5/EMMC_D2_u
GPIO2_D6/EMMC_D3_u
GPIO2_D7/EMMC_D4_u
GPIO3_C0/EMMC_D5_u
GPIO3_C1/EMMC_D6_u
GPIO3_C2/EMMC_D7_u
GPIO3_C3/EMMC_CMD_u
GPIO3_C5/EMMC_CLKOUT_u
GPIO3_C6/EMMC_PWREN_d
VCCIO2
R211
NC/10K
C58
100nF/10V
VCC18_EMMC
RK3328
BGA395_14R00X14R00X1R24
U2D
GPIO1_A0/SDMMC0_D0/UART2_TX_M0_u
GPIO1_A1/SDMMC0_D1/UART2_RX_M0_u
GPIO1_A2/SDMMC0_D2/JTAG_TCK_u
GPIO1_A3/SDMMC0_D3/JTAG_TMS_u
GPIO1_A4/SDMMC0_CMD_u
GPIO1_A6/SDMMC0_CLK/TEST_CLK0_d
GPIO1_A5/SDMMC0_DETN_u
VCCIO3
SDMMC0_D0 [9]
SDMMC0_D1 [9]
SDMMC0_D2/JTAG_TCK [9]
SDMMC0_D3/JTAG_TMS [9]
SDMMC0_CMD [9]
SDMMC0_CLK [9]
SDMMC0_DET [9]
VCC_IO_SD
C57
100nF/10V
RK3328
BGA395_14R00X14R00X1R24
U2M
DDR3/DDR4
DDR_DQ0
DDR_DQ1
DDR_DQ2
DDR_DQ3
DDR_DQ4
DDR_DQ5
DDR_DQ6
DDR_DQ7
DDR_DQM0
DDR_DQS0_P
DDR_DQS0_N
DDR_DQ8
DDR_DQ9
DDR_DQ10
DDR_DQ11
DDR_DQ12
DDR_DQ13
DDR_DQ14
DDR_DQ15
DDR_DQM1
DDR_DQS1_P
DDR_DQS1_N
DDR_DQ16
DDR_DQ17
DDR_DQ18
DDR_DQ19
DDR_DQ20
DDR_DQ21
DDR_DQ22
DDR_DQ23
DDR_DQM2
DDR_DQS2_P
DDR_DQS2_N
DDR_DQ24
DDR_DQ25
DDR_DQ26
DDR_DQ27
DDR_DQ28
DDR_DQ29
DDR_DQ30
DDR_DQ31
DDR_DQM3
DDR_DQS3_P
DDR_DQS3_N
DDR3_A0/DDR4_A10
DDR3_A1/DDR4_A9
DDR3_A2/DDR4_A4
DDR3_A3/DDR4_A6
DDR3_A4/DDR4_A5
DDR3_A5/DDR4_A8
DDR3_A6/DDR4_A7
DDR3_A7/DDR4_A11
DDR3_A8/DDR4_A13
DDR3_A9/DDR4_A0
DDR3_A10/DDR4_CS0n
DDR3_A11/DDR4_A3
DDR3_A12/DDR4_BA1
DDR3_A13/DDR4_A2
DDR3_A14/DDR4_A1
DDR3_A15/DDR4_ODT0
DDR3_BA0/DDR4_BG0
DDR3_BA1/DDR4_CASn/DDR4_A15
DDR3_BA2/DDR4_BA0
DDR3_CSN0/DDR4_ACTn
DDR3_CSN1/DDR4_CS1N
DDR3_ODT0/DDR4_Wen/DDR4_A14
DDR3_ODT1/DDR4_ODT1
DDR3_CLKP/DDR4_CLKP
DDR3_CLKN/DDR4_CLKN
DDR3_CKE/DDR4_RASn/DDR4_A16
DDR3_RASn/DDR4_CKE
DDR3_CASn/DDR4_A12
DDR3_WEn/DDR4_BG1
DDR3_RESETn/DDR4_RESETN
DDR4_A10 [8]
DDR4_A9 [8]
DDR4_A4 [8]
DDR4_A6 [8]
DDR4_A5 [8]
DDR4_A8 [8]
DDR4_A7 [8]
DDR4_A11 [8]
DDR4_A13 [8]
DDR4_A0 [8]
DDR4_CS0n [8]
DDR4_A3 [8]
DDR4_BA1 [8]
DDR4_A2 [8]
DDR4_A1 [8]
DDR4_ODT0 [8]
DDR4_BG0 [8]
DDR4_CASn/A15 [8]
DDR4_BA0 [8]
DDR4_ACTn [8]
DDR4_WEn/A14 [8]
DDR4_CLKP [8]
DDR4_CLKN [8]
DDR4_RASn/A16 [8]
DDR4_CKE [8]
DDR4_A12 [8]
DDR4_RST [8]
VDD_DDRIO
VCC_DDR
C52
C53
C54
C55
C56
10nF/50V
100nF/10V
100nF/10V
10uF/6.3V
10uF/6.3V
RK3328
BGA395_14R00X14R00X1R24
VCC_IO_33
L23
NC/180R-100M
L0402
VDD_5V
VCC_IO_SD
U20
IN
OUT
GND
EN
BP
TT2102
C239
4.7uF/10V
C0402
R206
10K
R207
124K/1%
C240
4.7uF/10V
C0402
R210
66.5K/1%
R208
100K/1%
R212
10K
Q5
L2N7002WT1G
R209
47K
Q10
L2N7002WT1G
R214
47K
R213
47K
[6] SDIO_PER
FRIENDLY ELEC
NanoPi R2S
Size A3
Document Number 05.DDR/eMMC/TF Controler
Rev 1912
Date: Tuesday, October 20, 2020
Sheet 5 of 14

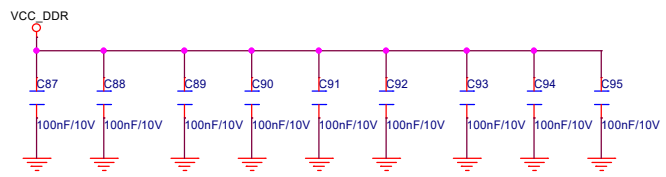

U2E
GPIO1_B0/UART0_RX/GMAC_TXD1_M1_u
GPIO1_B1/UART0_TX/GMAC_TXD0_M1_u
GPIO1_B2/UART0_RTSN/GMAC_RXD1_M1_d
GPIO1_B3/UART0_CTSN/GMAC_RXD0_M1_d
GPIO1_B4/SDMMC1_CLK/GMAC_TXCLK_M1_d
GPIO1_B5/SDMMC1_CMD/GMAC_RXCLK_M1_u
GPIO1_B6/SDMMC1_D0/GMAC_RXD3_M1_u
GPIO1_B7/SDMMC1_D1/GMAC_RXD2_M1_u
GPIO1_C0/SDMMC1_D2/GMAC_TXD3_M1_u
GPIO1_C1/SDMMC1_D3/GMAC_TXD2_M1_u
GPIO1_C2/SDMMC1_PWREN/GMAC_CRS_M1_d
GPIO1_C3/SDMMC1_DET/GMAC_MDIO_M1/PDM_FSYNC_M1_u
GPIO1_C5/I2S2_MCLK/GMAC_CLK_M1_d
GPIO1_C6/I2S2_SCLK_M0/GMAC_RXDV_M1/PDM_CLK_M1_u
GPIO1_C7/I2S2_LRCK_TX_M0/GMAC_MDC_M1/PDM_SDI0_M1_d
GPIO1_D2/I2S2_LRCK_RX_M0/CLKOUT_GMAC_M2/PDM_SDI3_M1_d
GPIO1_D0/I2S2_SDI_M0/GMAC_RXER_M1/PDM_SDI1_M1_d
GPIO1_D1/I2S2_SDO_M0/GMAC_TXEN_M1/PDM_SDI2_M1_d
GPIO1_D4/CLK32KOUT_M1_d
VCCIO4
Y15
AA15
W14
V13
AA20
Y20
W19
W18
AA19
Y19
W20
Y21
W15
W16
W17
V12
AA17
Y17
V14
U12
GMAC_TXD1 [13]
GMAC_TXD0 [13]
GMAC_RXD1 [13]
GMAC_RXD0 [13]
GMAC_TXCLK [13]
GMAC_RXCLK [13]
GMAC_RXD3 [13]
GMAC_RXD2 [13]
GMAC_TXD3 [13]
GMAC_TXD2 [13]
PHY-RESET [13]
GMAC_MDIO [13]
GMAC_CLK [13]
GMAC_RXDV [13]
GMAC_MDC [13]
USB20_HOST_DRV [10]
PMIC_INT [3]
GMAC_TXEN [13]
SDIO_PER [5]
VDD_PHY_IO
C59
100nF/10V
RK3328
BGA395_14R00X14R00X1R24
U2K
USB2.0
USB0_DM
USB0_DP
USB20_VBUS
USB1_DM
USB1_DP
USB20_EXTR
USB20_AVDD_3V3
USB20_AVDD_1V8
USB20_DVDD_1V0
Y6
AA6
T6
Y7
AA7
W6
T5
U5
V5
OTG20_DM [10]
OTG20_DP [10]
OTG20_ID [10]
USB1_DM [10]
USB1_DP [10]
R25
133R/1%
R0201
VCC_IO_33
C63
100nF/10V
USB20/FEPHY_1V8
C64
100nF/10V
C65
1uF/6.3V
VCC_18
L10
180R-100M
L0402
C66
4.7uF/10V
C0402
USB20_DVDD_1V0
C67
100nF/10V
C68
1uF/6.3V
R142
NC/1R
R0402
VDD_10
C69
4.7uF/10V
C0402
R129
1R
R0402
VDD_LOG
RK3328
BGA395_14R00X14R00X1R24
U2J
10/100M Ethernet PHY
FEPHY_TXP
FEPHY_TXN
FEPHY_RXP
FEPHY_RXN
FEPHY_EXTRES
FEPHY_AVDD_1V8
(TSADC_1V8)
FEPHY_AVDD_1V0
Y9
AA9
Y8
AA8
V7
U6
V6
R23
6.49K/1%
R0201
USB20/FEPHY_1V8
C60
100nF/10V
FEPHY_AVDD_1V0
内置FEPHY不用时，这两路电源也需要供电，
内部TSADC和FEPHY电源共用，
如果不供电，会造成TSADC无法工作。
C61
100nF/10V
VDD_10
L11
180R-100M
L0402
VDD_LOG
L12
NC/180R-100M
L0402
C62
4.7uF/10V
C0402
Embed FEPHY
U2G
GPIO3_A4/TSP_D0/CIF_D0/SDMMC0EXT_D0/UART1_TX/USB3PHY_DEBUG4_u
GPIO3_A5/TSP_D1/CIF_D1/SDMMC0EXT_D1/UART1_RTSN/USB3PHY_DEBUG5_u
GPIO3_A6/TSP_D2/CIF_D2/SDMMC0EXT_D2/UART1_RX/USB3PHY_DEBUG6_u
GPIO3_A7/TSP_D3/CIF_D3/SDMMC0EXT_D3/UART1_CTSN/USB3PHY_DEBUG7_u
GPIO3_A0/TSP_VALID/CIF_VSYNC/SDMMC0EXT_CMD/SPI_CLK_M2/USB3PHY_DEBUG1/I2S2_SCLK_M1_u
GPIO3_A2/TSP_CLK/CIF_CLKIN/SDMMC0EXT_CLK/SPI_RXD_M2/USB3PHY_DEBUG3/I2S2_SDI_M1_d
GPIO3_A1/TSP_FAIL/CIF_HREF/SDMMC0EXT_DET/SPI_TXD_M2/USB3PHY_DEBUG2/I2S2_SDO_M1_u
GPIO3_B0/TSP_D4/CIF_D4/SPI_CSN0_M2/I2S2_LRCK_TX_M1/USB3PHY_DEBUG8/I2S2_LRCK_RX_M1_d
VCCIO6
E2
D1
F2
F1
E3
E1
D2
F3
H6
UART1_TX/GPIO3_A4 [9]
UART1_RX/GPIOG3_A6 [9]
GPIO3_A0/SPI_CLK [9]
GPIO3_A2/SPI-RXD [9]
GPIO3_A1/SPI-TXD [9]
GPIO3_B0/SPI_CSN0 [9]
VCC_IO_33
RK3328
BGA395_14R00X14R00X1R24
U2L
USB3.0
USB30_RXP
USB30_RXN
USB30_TXP
USB30_TXN
USB30_DM
USB30_DP
USB30_VBUS
USB30_EXTR
USB30_AVDD_3V3
USB30_AVDD_1V8
USB30_DVDD_1V0
J2
J3
H3
H2
K2
K1
K7
G5
J6
J5
J7
USB30_RXP [12]
USB30_RXN [12]
USB30_TXP [12]
USB30_TXN [12]
USB30_DM [12]
USB30_DP [12]
VCC_IO_33
R24
2K/1%
VCC_IO_33
VDDPLL/USB30_1V8
VDDPLL/USB30_1V0
C70
100nF/10V
C71
100nF/10V
C72
100nF/10V
RK3328
BGA395_14R00X14R00X1R24
FRIENDLY ELEC
NanoPi R2S
Size A3
Document Number 06.FLASH/TF/USB Controler
Rev 1912
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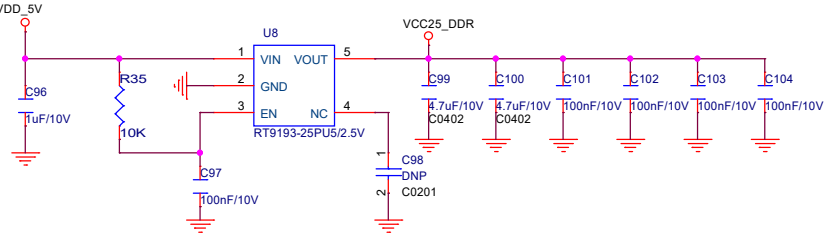


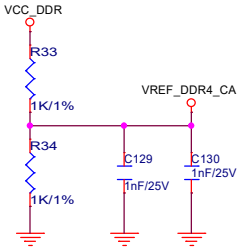
FRIENDLY ELEC

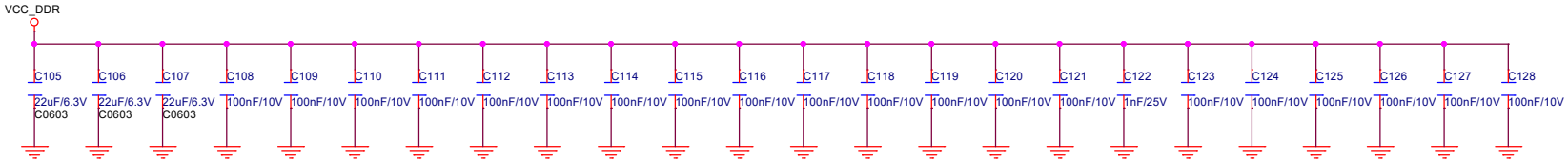
NanoPi R2S		
Size A3	Document Number 07.HDMI/AV/EPHY/GPIO Controler	Rev 1912
Date: Monday, September 21, 2020	Sheet 7 of 14	





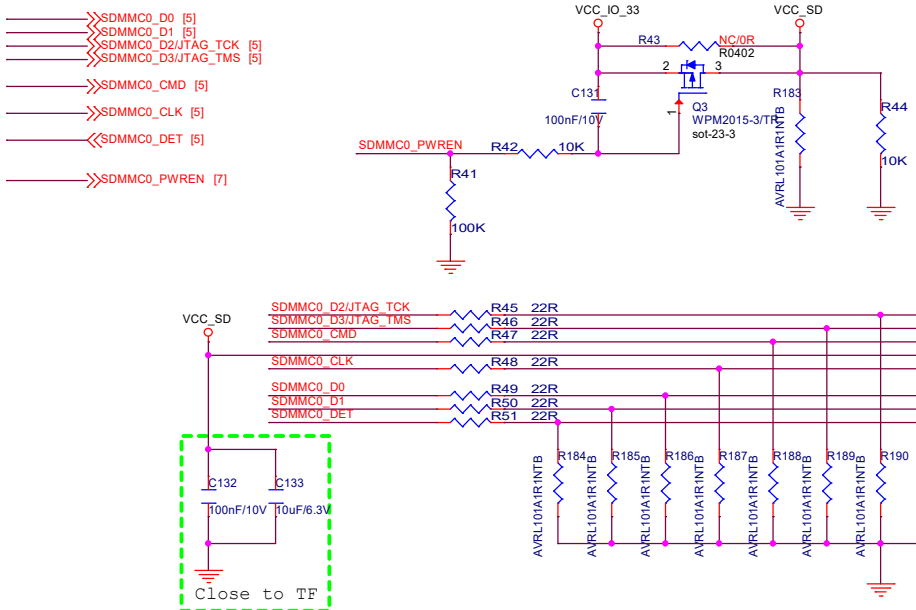


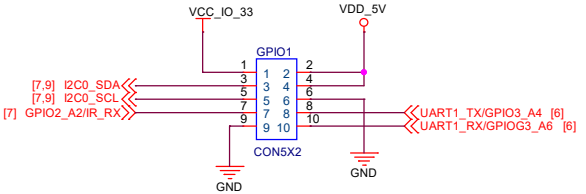




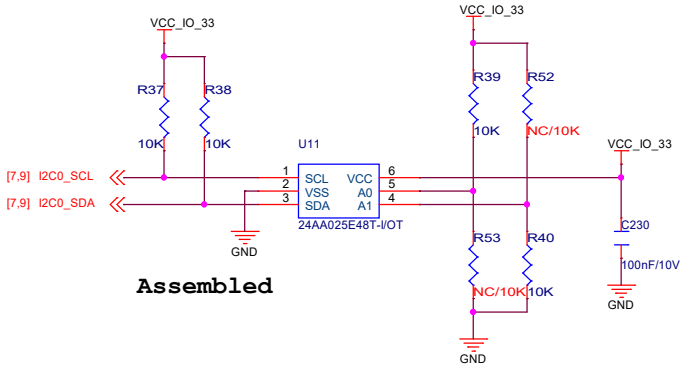
FRIENDLY ELEC

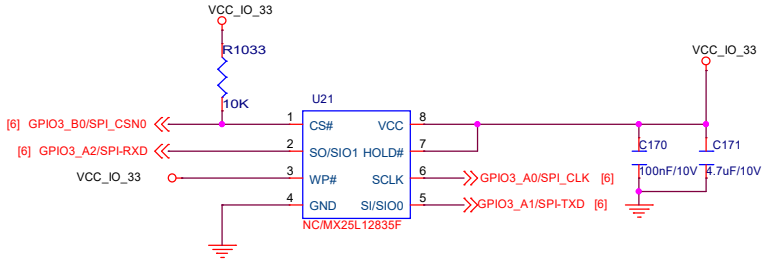
NanoPi R2S		
Size A3	Document Number 08.RAM DDR4 2x16bit	Rev 1912
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EEPROM with MAC Address

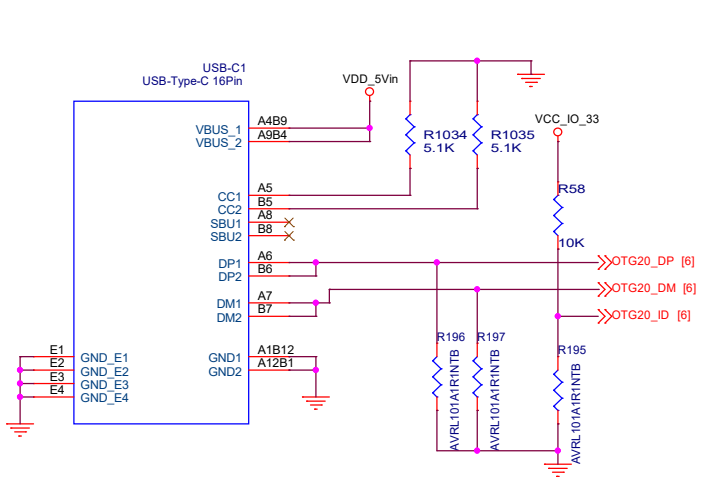


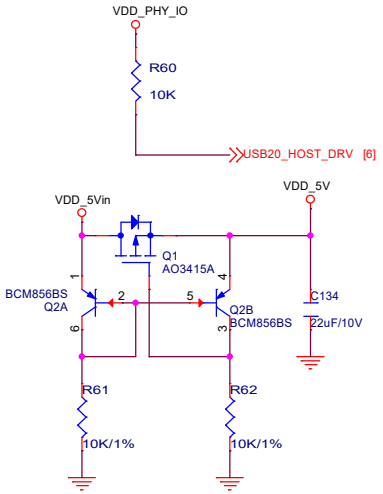


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NanoPi R2S		
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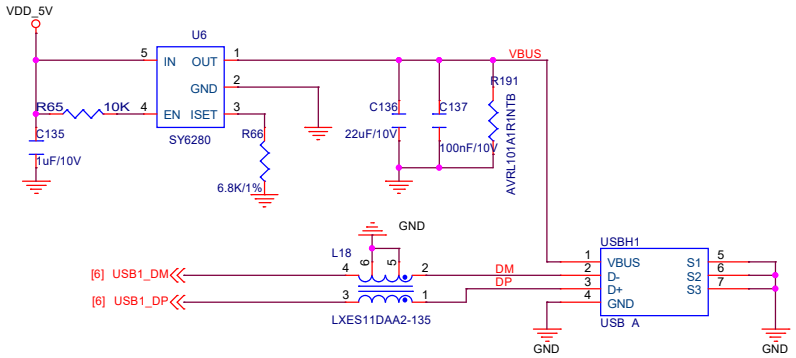
MicroUSB:PWR and Slave



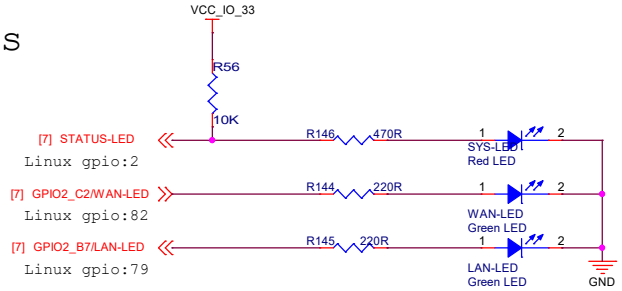




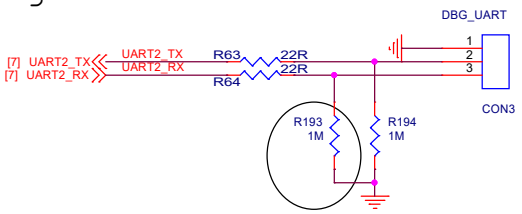
USB Host



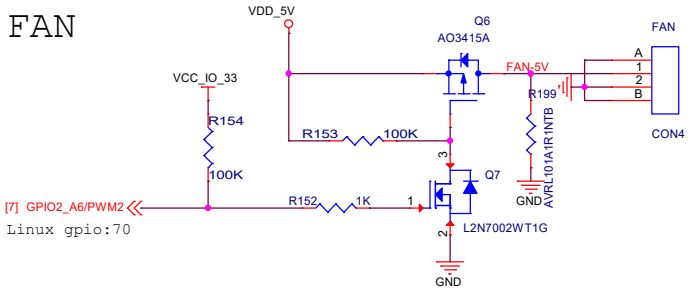
LEDs



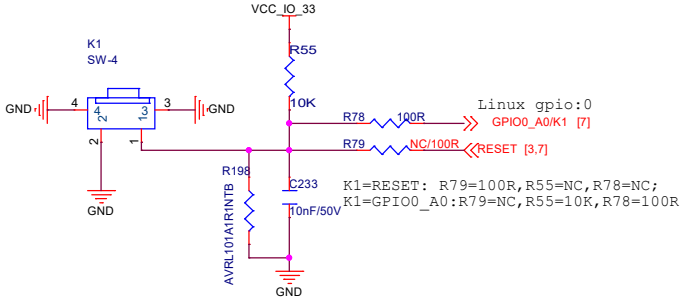
Debug UART2



FAN

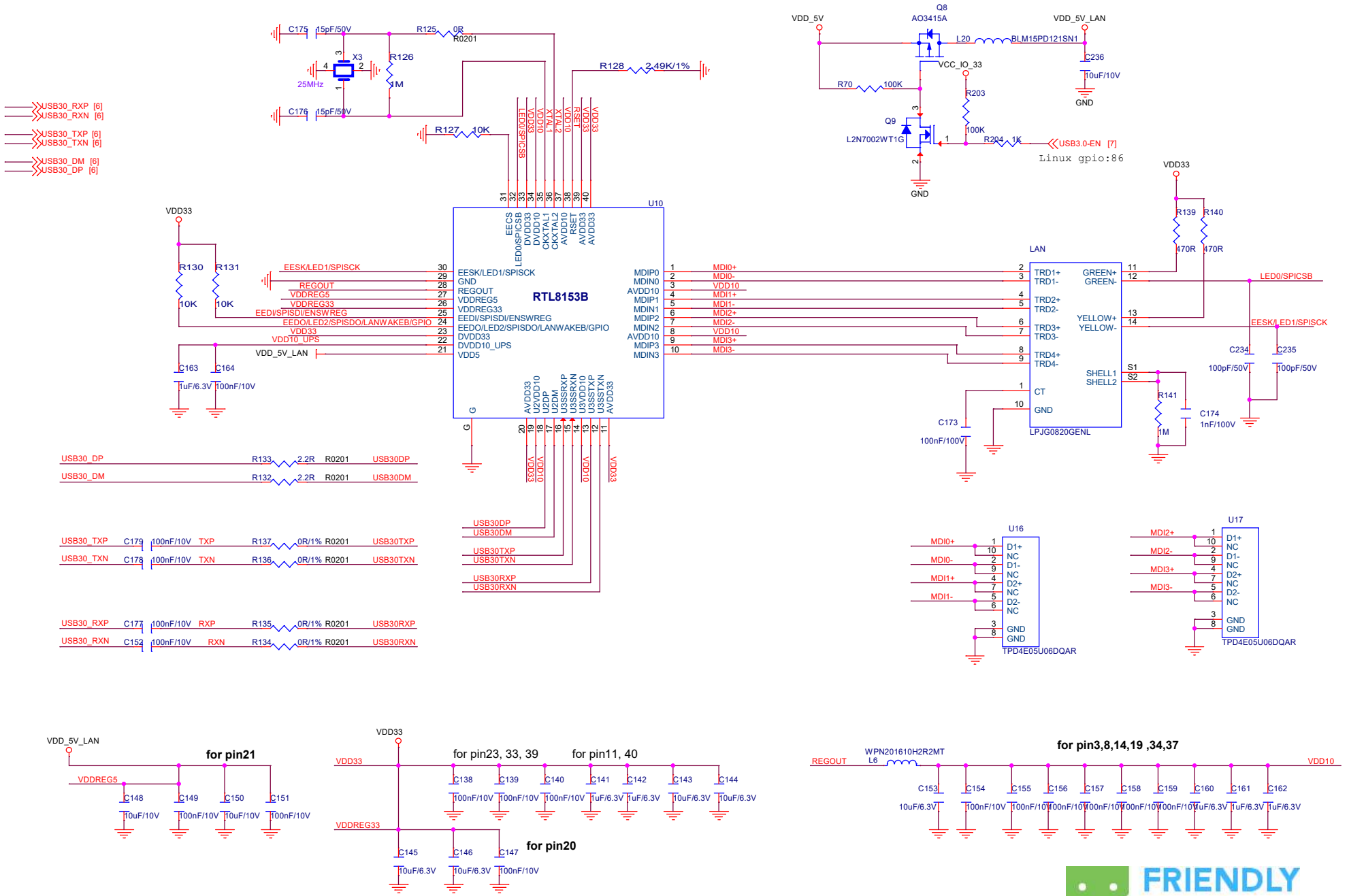


KEY



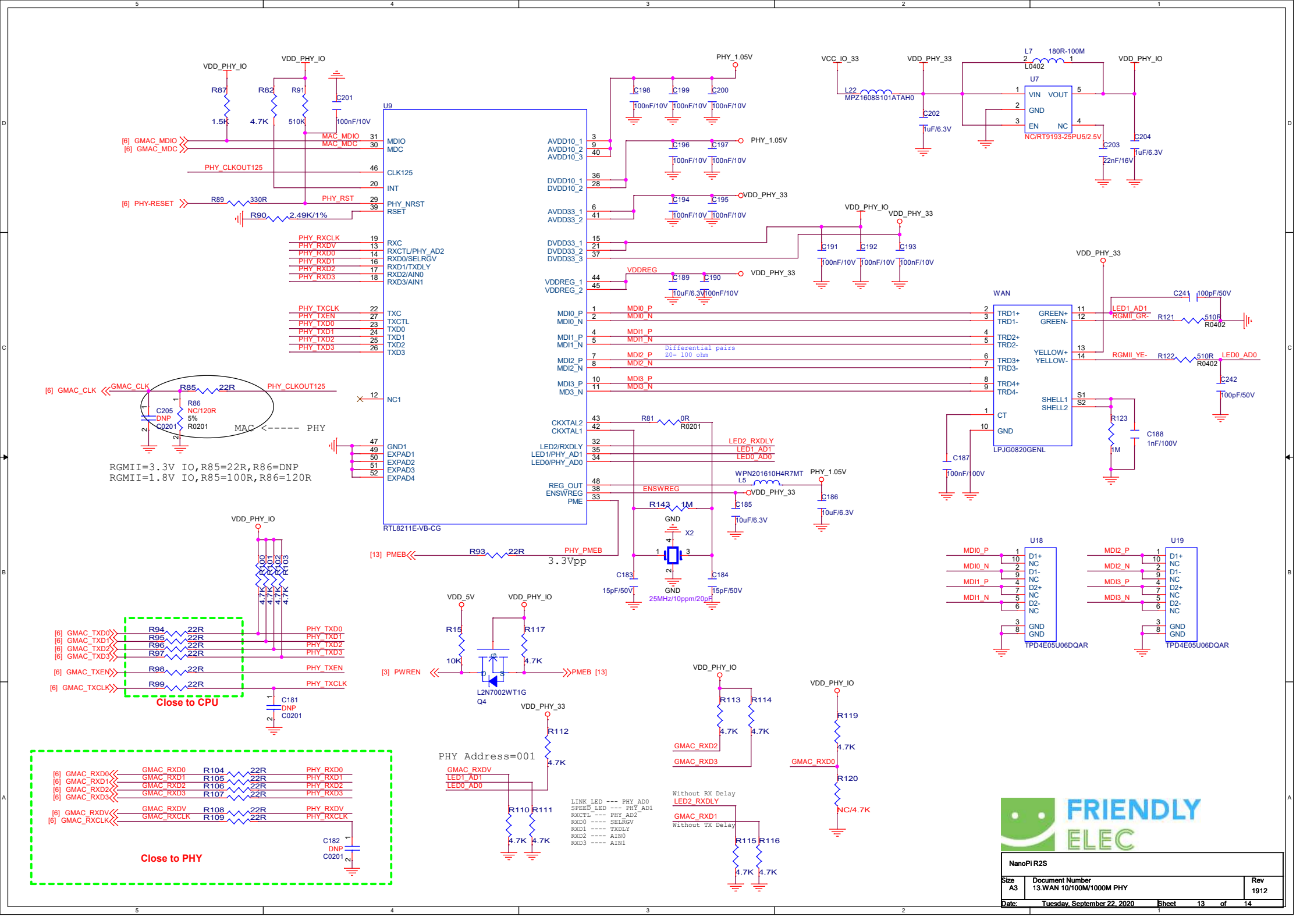
FRIENDLY ELEC

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NanoPi R2S		
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U9
RTL8211E-VB-CG
MDIO
MDC
CLK125
INT
PHY_NRST
RSET
RXC
RXCTL/PHY_AD2
RXD0/SELRGV
RXD1/TXDLY
RXD2/AIN0
RXD3/AIN1
TXC
TXCTL
TXD0
TXD1
TXD2
TXD3
NC1
GND1
EXPAD1
EXPAD2
EXPAD3
EXPAD4
AVDD10_1
AVDD10_2
AVDD10_3
DVDD10_1
DVDD10_2
AVDD33_1
AVDD33_2
DVDD33_1
DVDD33_2
DVDD33_3
VDDREG_1
VDDREG_2
MDI0_P
MDI0_N
MDI1_P
MDI1_N
MDI2_P
MDI2_N
MDI3_P
MDI3_N
CKXTAL2
CKXTAL1
LED2/RXDLY
LED1/PHY_AD1
LED0/PHY_AD0
REG_OUT
ENSWREG
PME
Differential pairs
Z0= 100 ohm
WAN
LPJG0820GENL
TRD1+
TRD1-
TRD2+
TRD2-
TRD3+
TRD3-
TRD4+
TRD4-
CT
GND
GREEN+
GREEN-
YELLOW+
YELLOW-
SHELL1
SHELL2
U7
NC/RT9193-25PU5/2.5V
VIN
VOUT
GND
EN
NC
L7
180R-100M
L0402
L22
MPZ1608S101ATAH0
U18
U19
TPD4E05U06DQAR
L2N7002WT1G
Q4
WPN201610H4R7MT
L5
25MHz/10ppm/20pF
3.3Vpp
PHY Address=001
LINK_LED --- PHY_AD0
SPEED_LED --- PHY_AD1
RXCTL --- PHY_AD2
RXD0 --- SELRGV
RXD1 --- TXDLY
RXD2 --- AIN0
RXD3 --- AIN1
Without RX Delay
Without TX Delay
Close to CPU
Close to PHY
RGMII=3.3V IO,R85=22R,R86=DNP
RGMII=1.8V IO,R85=100R,R86=120R
MAC <----- PHY
VDD_PHY_IO
VDD_PHY_33
PHY_1.05V
VCC_IO_33
VDD_5V
PHY_RXCLK
PHY_RXDV
PHY_RXD0
PHY_RXD1
PHY_RXD2
PHY_RXD3
PHY_TXCLK
PHY_TXEN
PHY_TXD0
PHY_TXD1
PHY_TXD2
PHY_TXD3
PHY_CLKOUT125
PHY_RST
PHY_PMEB
PHY_TXD0
PHY_TXD1
PHY_TXD2
PHY_TXD3
PHY_TXEN
PHY_TXCLK
PHY_RXD0
PHY_RXD1
PHY_RXD2
PHY_RXD3
PHY_RXDV
PHY_RXCLK
LED1_AD1
LED0_AD0
LED2_RXDLY
RGMII_GR-
RGMII_YE-
GMAC_MDIO
GMAC_MDC
GMAC_CLK
GMAC_TXD0
GMAC_TXD1
GMAC_TXD2
GMAC_TXD3
GMAC_TXEN
GMAC_TXCLK
GMAC_RXD0
GMAC_RXD1
GMAC_RXD2
GMAC_RXD3
GMAC_RXDV
GMAC_RXCLK
PWREN
PMEB
ENSWREG
PHY-RESET
R87 1.5K
R82 4.7K
R91 510K
R89 330R
R90 2.49K/1%
R85 22R
R86 NC/120R 5%
R81 0R
R143 1M
R93 22R
R94 22R
R95 22R
R96 22R
R97 22R
R98 22R
R99 22R
R100 4.7K
R101 4.7K
R102 4.7K
R103 4.7K
R104 22R
R105 22R
R106 22R
R107 22R
R108 22R
R109 22R
R15 10K
R117 4.7K
R112 4.7K
R110 4.7K
R111 4.7K
R113 4.7K
R114 4.7K
R119 4.7K
R120 NC/4.7K
R115 4.7K
R116 4.7K
R121 510R
R122 510R
R123 1M
C201 100nF/10V
C202 1uF/6.3V
C203 22nF/16V
C204 1uF/6.3V
C205 DNP C0201
C181 DNP C0201
C182 DNP C0201
C183 15pF/50V
C184 15pF/50V
C185 10uF/6.3V
C186 10uF/6.3V
C187 100nF/100V
C188 1nF/100V
C189 10uF/6.3V
C190 100nF/10V
C191 100nF/10V
C192 100nF/10V
C193 100nF/10V
C194 100nF/10V
C195 100nF/10V
C196 100nF/10V
C197 100nF/10V
C198 100nF/10V
C199 100nF/10V
C200 100nF/10V
C241 100pF/50V
C242 100pF/50V
R0402
R0201
X2
GND
FRIENDLY ELEC
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